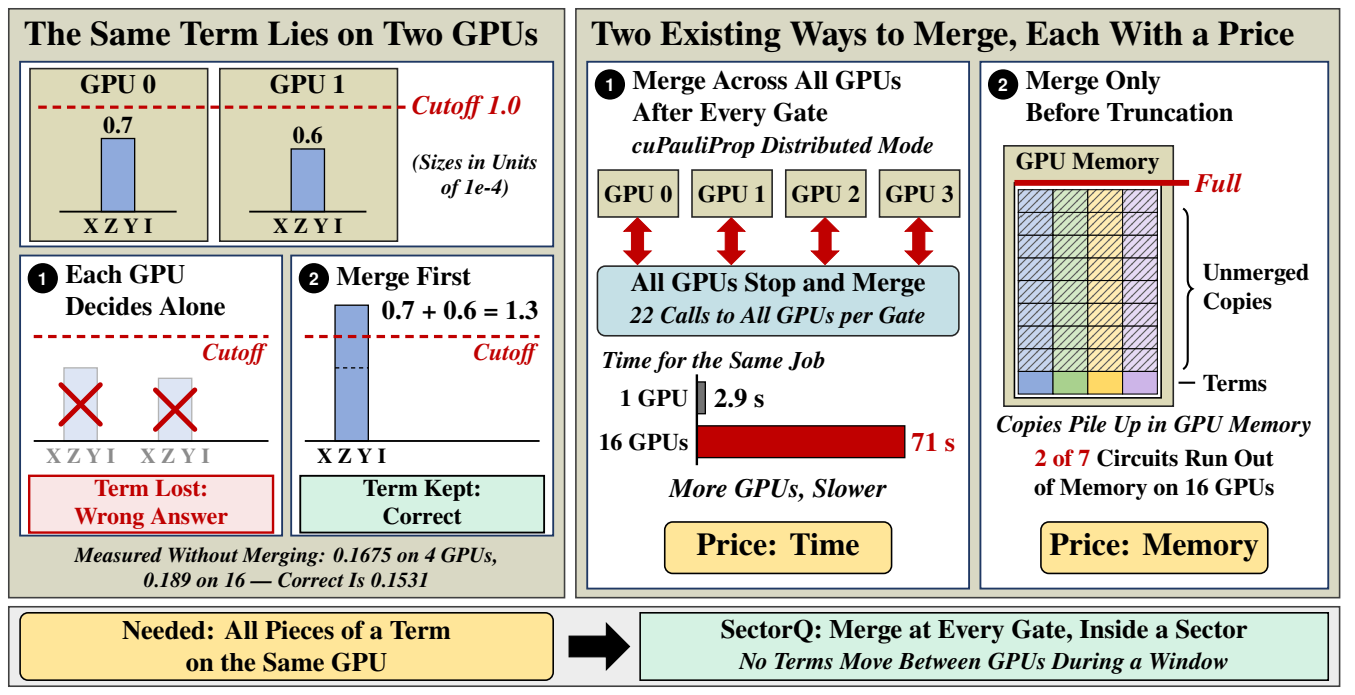

The Same Term Lies on Two GPUs
GPU 0
GPU 1
Cutoff 1.0
0.7
0.6
X Z Y I
X Z Y I
(Sizes in Units of 1e-4)
1 Each GPU Decides Alone
Cutoff
X Z Y I
X Z Y I
Term Lost: Wrong Answer
2 Merge First
0.7 + 0.6 = 1.3
Cutoff
X Z Y I
Term Kept: Correct
Measured Without Merging: 0.1675 on 4 GPUs, 0.189 on 16 — Correct Is 0.1531
Two Existing Ways to Merge, Each With a Price
1 Merge Across All GPUs After Every Gate
cuPauliProp Distributed Mode
GPU 0
GPU 1
GPU 2
GPU 3
All GPUs Stop and Merge
22 Calls to All GPUs per Gate
Time for the Same Job
1 GPU
2.9 s
16 GPUs
71 s
More GPUs, Slower
Price: Time
2 Merge Only Before Truncation
GPU Memory
Full
Unmerged Copies
Terms
Copies Pile Up in GPU Memory
2 of 7 Circuits Run Out of Memory on 16 GPUs
Price: Memory
Needed: All Pieces of a Term on the Same GPU
SectorQ: Merge at Every Gate, Inside a Sector
No Terms Move Between GPUs During a Window